Assignment 4

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Code:
Interactive Weather Dashboard using Charts

1. HTML Structure

```
<header>
  <div class="header-top">
    <h1>Weather Dashboard</h1>

    <div class="search-box">
      <input type="text" id="locationInput" placeholder="Enter city...">
      <button id="searchBtn">Search</button>
    </div>
  </div>

  <div class="location-datetime-info">
    <h2>Weather for: <span id="locationDisplay">Pune, India</span></h2>
    <p id="datetimeDisplay"></p>
  </div>
</header>

<main class="dashboard-container">

  <div class="chart-card">
    <h2>Temperature</h2>
    <canvas id="temperatureChart"></canvas>
  </div>

  <div class="chart-card">
    <h2>Humidity</h2>
    <canvas id="humidityChart"></canvas>
  </div>

  <div class="chart-card">
    <h2>Wind Speed</h2>
    <canvas id="windChart"></canvas>
  </div>

</main>
```

2. CSS Styling

```
body{
background: linear-gradient(135deg, #0f2027, #203a43, #2c5364);
color:white;
}

.chart-card{
background:rgba(255,255,255,0.05);
backdrop-filter:blur(15px);
border-radius:15px;
padding:20px;
transition:0.3s;
}

.chart-card:hover{
transform:translateY(-10px);
box-shadow:0 0 20px rgba(0,224,255,0.4);
}

.dashboard-container{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(300px,1fr));
gap:20px;
}
```

3. JavaScript Functionality

```
function randomData(min,max){
return Array.from({length:7},()=>Math.floor(Math.random()*(max-min+1)+min));
}

searchBtn.addEventListener("click",()=>{
let city = input.value;
locationDisplay.innerText = city;

tempChart.data.datasets[0].data = randomData(20,35);
humChart.data.datasets[0].data = randomData(40,90);
windChart.data.datasets[0].data = randomData(5,25);

tempChart.update();
humChart.update();
windChart.update();
});
```